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DIRECT TRANSFER OF PROSODY IN SPEECH-TO-SPEECH TRANSLATION USING DISENTANGLED SPEECH TOKENS

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ABSTRACT

Current speech-to-speech translation (S2ST) systems often fail to preserve vital prosodic nuances, the rhythm, pitch, and emotion, of the original speaker. This failure stems from the lack of prosodically parallel data, forcing systems into indirect modeling and creating a significant training-inference mismatch. In this work, we overcome this fundamental challenge with PASST (Prosody-Aligned Speech-to-Speech Translation), a novel framework for direct prosody transfer in speech-to-speech translation. We first disentangle speech into its core components: linguistic content, prosody, and timbre. We then selectively translate only the linguistic content while preserving the other aspects of source speech. Our key insight is to leverage the attention maps from the content translation module to precisely align the source prosody and acoustic features with the translated content. Experiments demonstrate that PASST significantly outperforms baselines in prosody preservation. More importantly, this faithful transfer of rich prosodic detail results in translated speech with better naturalness.

Index Terms— speech-to-speech translation, prosody transfer, disentangled codec

1. INTRODUCTION

The goal of speech-to-speech translation (S2ST) is to enable seamless, natural communication across linguistic barriers. Traditional cascaded systems—which chain together automatic speech recognition (ASR), machine translation (MT), and text-to-speech (TTS) models—fall short of this goal. Recently, the field has shifted towards direct S2ST models that learn an end-to-end mapping from source to target speech [1–5]. This unified approach prevents information loss and error propagation, making more efficient and authentic translation possible. The architecture of direct S2ST has evolved rapidly. Early models operated on continuous features like log-mel spectrograms using sequence-to-sequence architectures [1, 2], but the advent of self-supervised learning (SSL) [6, 7] introduced a powerful new paradigm: enabling the use of powerful transformer architectures originally designed for text translation [3–5] adopted to process discrete SSL features of speech. These discrete features primarily encode content information, simplifying translation by filtering out excessive acoustic details and enabling the utilization of only semantically parallel data. Recently, state-of-the-art speech codecs [8, 9] encoding speech into discrete tokens have begun replacing SSL-based features.

Human speech conveys meaning beyond semantic content; its meaning is profoundly shaped by prosody, the intonation, rhythm, and emphasis that convey intent, emotion, and nuance. However, these vital dimensions are either ignored by state-of-the-art S2ST systems [3–5] or modeled inadequately. Some methods attempt prosody modeling by conditioning speech generation with prosodic features [9, 10], but due to the scarcity of prosodically parallel training data, they rely on indirect formulations without any explicit prosody transfer, and often suffer from a training-inference mismatch. These approaches fail to preserve rich prosodic details, resulting in unnatural translations lacking expressiveness.

We introduce PASST (Prosody-Aligned Speech-to-Speech Translation), a novel framework that fundamentally rethinks prosody preservation. Our approach leverages a disentangled speech codec, FACodec [11], to separate speech into its core components: linguistic content, prosody, acoustic details, and timbre. The key to our method is selective translation: we translate only the linguistic content tokens, while preserving all other features from the source speech. The foundation of our method lies in using the internal cross-attention maps of the translation model to create a precise, fine-grained alignment between the original source prosody and the newly generated target content. This enables the direct transfer of local prosodic cues much more effectively and accurately. Our experiments demonstrate that PASST achieves significant improvements in both prosody similarity and overall naturalness compared to indirect modeling of prosody.

2. RELATED WORK

Most S2ST methods, based on sequence-to-sequence translation of semantic speech units, ignore prosodic aspects of source speech [3–5, 12]. Some approaches [9, 10, 13] attempt

s2ut-based

Speech samples: https://lec-synt.github.io/passt-demo/

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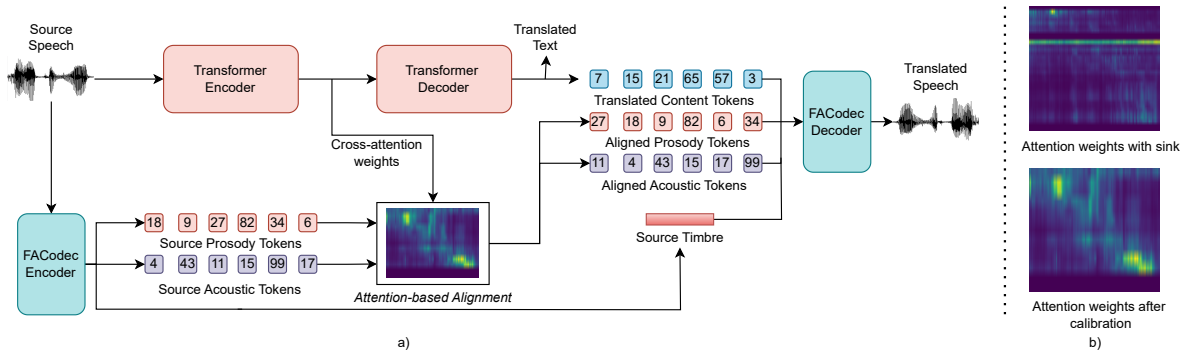


Fig. 1. a) Proposed S2ST which translates content with a content translator module, and transfer other aspects of source speech by aligning them to the translated content utilizing the attention maps of translation. b) The effect of attention calibration

to incorporate prosody by conditioning speech generation with encoded prosodic features alongside translated semantic content. However, prosodically parallel data, where translation pairs share both meaning and prosody, is extremely rare. Consequently, during training, prosody-aware S2ST systems use prosody features from target speech when generating the target speech itself, while at inference, prosody features come from source speech in a different language. This training-inference mismatch and lack of explicit prosody transfer cause these methods to fail at preserving source prosody.

Research shows that fundamental prosodic patterns such as intonation, emphasis remain remarkably similar across certain language pairs, despite changes in words and sequential order. Studies have shown strong prosodic correlations between English-Portuguese [14], English-Spanish [15], and across English-Spanish-German in professional dubbing [16]. These findings provide strong evidence that core prosodic patterns can be directly transferred between languages and this principle of cross-lingual prosodic correspondence forms the foundation for our method.

3. METHODOLOGY

The proposed method uses a speech codec to disentangle speech into aligned tokens for linguistic content, prosody, acoustic detail, and timbre. An encoder–decoder transformer translates the source content tokens into target content tokens. At inference, source non-content tokens are aligned to the translated content using attention maps, which capture fine-grained correspondences between source speech and target content tokens. The codec decoder then reconstructs translated speech, preserving source prosody and acoustic detail. This direct alignment enables more faithful prosody transfer than conventional indirect approaches. The overview of the proposed method can be seen in Figure 1.

3.1. Distentangled Speech Codec

We employ FACodec [11], proposed in NaturalSpeech 3, as our speech codec. FACodec encodes speech into discrete tokens using residual vector quantization (RVQ). Unlike other speech codecs, FACodec applies disentanglement to obtain tokens specific to different speech aspects: linguistic content, prosody, acoustic detail, and timbre, namely z_c, z_p, z_a, h. FACodec achieves disentanglement by imposing additional specific target and adversarial objectives on different token types beyond the speech reconstruction objective. All tokens are extracted every 12.5 ms and inherently aligned, resulting in fine-grained representations. The FACodec decoder D_{FA} reconstructs speech as $\tilde{x} = D_{FA}(z_c, z_p, z_a, h)$. In this work we discarded second-level content tokens of RVQ as most of the information resides in first-level content tokens.

3.2. Content Translation Module

The content translator module responsibles for converting the linguistic content of the source speech into a sequence of target language content tokens. Our architecture, inspired by TransVIP [9], is composed of three key components: a Speech Semantic Encoder to extract meaning, an Isochrony Encoder to capture temporal dynamics, and a two-stage autoregressive decoder to generate the translated output.

Speech Semantic Encoder: As speech semantic encoder, we employ an conformed based SSL model, w2v-BERT 2.0 [7] which is pre-trained on large-scale multi-lingual data with constrastive learning and masked prediction objective. The semantic encoder E_s takes mel-spectrogram of source speech x as input and encodes them into frame-level latent representations $E_s(x) = h_s = [h_1, ..., h_N]$ with a 160 ms frame-length.

Isochrony Encoder: Similarly to TransVIP, we encode duration information from source speech using an isochrony encoder to guide the translation according to the source duration. Isochrony encoder E_{iso} is an embedding layer that embeds and sums regular and reverse positional indices along with the binary speech/non-speech indicator vector.

Decoder: Decoder D is an autoregressive transformer decoder, which takes speech semantic encoder and isochrony encoder outputs which are concatenated, namely h_s, i as input and predicts first the translated text sequence T^{target} followed by FACodec content sequence z_c^{target}. The decoding can be formulated as

$$T_t^{target} = D(T_{<t}^{target} | h_s, i) \quad (1)$$

$$z_{c,t}^{target} = D(T^{target}, z_{c<t}^{target} | h_s, i) \quad (2)$$

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The sequential decoding of text followed by content tokens are mainly for guiding the speech decoding towards more robust prediction of translated semantic content.

3.3. Prosody Transfer with Explicit Alignment

The core idea of our work is rather the prediction of target prosody, to directly transfer the fine-grained prosodic and acoustic details from the source speech. The primary challenge lies in the sequential mismatch: the source prosody tokens $z_p^{source}, z_a^{source}$ are aligned with the source content, while our goal is to align them with the newly generated target content (z_c^{target}), which has a different sequential structure.

To solve this, we leverage the implicit alignment naturally learned by the cross-attention mechanism within the Transformer encoder-decoder [17]. The attention weights entails correspondence between input and output content, implying which parts of the source speech were most influential in generating each target token. By utilizing this information, we can create a precise mapping to transfer prosodic events in source speech, such as emphasis and intonation, onto the corresponding translated content.

Cross-attention weights between decoder and encoder outputs can be expressed as $A(h_s, z_c^{target}[i])$, where h_s represents the speech semantic encoder outputs and $z_c^{target}[i]$ is the ith predicted decoder output. To determine the most attended source speech region, we adopt two approaches: hard selection and soft selection.

Hard alignment. This approach deterministically selects the single source encoder feature $h_s[i]$ that receives the maximum attention weight when predicting $z_c^{target}[i]$. This method is simple and directly maps each target token to its most likely source origin.

$$h_s^{selected}[i] = h_s[argmax(A(h_s, z_c^{target}[i]))] \quad (3)$$

Soft Alignment. To create a potentially more robust alignment that is less sensitive to noisy individual attention scores, we also explore a soft, weighted approach. Here, the selected source representation is a weighted average of all source encoder features, using the normalized attention scores as weights:

$$h_s^{selected}[i] = \sum_j^T softmax(A(h_s, z_c^{target}[i]))h_s[j]. \quad (4)$$

The semantic encoder features have a frame rate of 160 ms, whereas the FACodec prosody and acoustic tokens have a much finer resolution of 12.5 ms. To resolve this temporal mismatch, for each target content token $z_c^{target}[i]$, we first identify its corresponding source time window using the selected feature $h_s^{selected}[i]$. We then perform average pooling on the high-resolution source prosody and acoustic token embeddings in that time window. This aggregates the relevant source token embeddings, producing a single aligned prosody $z_p^{aligned}[i]$ and acoustic embedding $z_a^{aligned}[i]$ for each target token. Finally, the FACodec decoder synthesizes the translated speech x^{target} using the translated content, the aligned prosody and acoustics, and the original source timbre vector : $x^{target} = D_{FA}(z_c^{target}, z_p^{aligned}, z_a^{aligned}, h)$.

Attention weight calibration. During initial experiments, we observed the "attention sink" phenomenon [18], where attention weights disproportionately collapse onto a few source tokens, often corresponding to non-speech regions. We identified that these sink tokens consistently exhibited a very small value vector norm $||V||$ in the attention calculation. We further identified those sink tokens are mostly non-speech regions in source speech. To mitigate this and ensure more meaningful alignments, we introduce a two-stage attention calibration method applied before the alignment step: 1) nullify attention weight to $\alpha_i = 0$ source token $h_{s,i}$ if source token is in non-speech region 2) nullify attention weight $\alpha_i = 0$ if source token value norm $||V_i||$ is smaller than a predefined threshold $||V_i|| < \beta$ where the threshold is half of the mean value norm calculated over all tokens $\beta = mean(||V_j||)/2$.

Text-based Alignment: We also experimented with machine translation alignment in the text domain. Using ASR-generated transcriptions and timestamps [19], we apply a MT aligner [20] to align source and translated words in speech, then transfer source prosody and acoustic token sequences to the corresponding target words with linear interpolation to match target content length. This allows comparison of our token-based alignment against robust text-based alignment.

4. EXPERIMENTS

We evaluate our proposed model on speech-to-speech translation for two language directions: English-to-French (En-Fr) and English-to-Spanish (En-Es).

4.1. Dataset

Our models are trained on the En-Fr and En-Es subsets of the CVSS-T dataset [21]. To address the much smaller size of the En-Es subset and ensure robust model training, we augment it with additional parallel data from the SeamlessAlign [22].

For evaluation, we selected the mExpresso dataset [23], a benchmark specifically designed for expressive speech. Derived from the Expresso corpus [24], mExpresso contains rich and varied prosodic details, making it ideal to assess prosody preservation. From the En-Fr and En-Es subsets, we curated a test set of 300 utterances from two speakers, ensuring coverage of the diverse prosodic styles.

4.2. Training & Architecture Details

Our architecture is adapted from the TransVIP [9] framework. The speech semantic encoder and autoregressive decoder follows the architecture of the SeamlessM4T-medium model [22], and we initialize their weights from the publicly available pre-trained checkpoints. To accommodate the generation of discrete audio tokens, we extend the decoder's vocabulary with embeddings for the FACodec content tokens; and these new embeddings are randomly initialized. The Isochrony Encoder follows the original TransVIP design.

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Table 1. S2ST results on mExpresso En-Fr

Methods	ASR-BLEU ↑	S-Sim ↑	A.PCP ↑	P-Sim ↑	NISQA ↑
Unit-to-Unit [5]	15.74	0.05	2.55	0.25	4.12
SeamlessM4T [22]	16.67	0.06	2.67	0.26	3.13
TransVIP [9]	16.35	0.18	2.71	0.32	3.43
PASST w/ hard align	18.17	0.23	2.95	0.37	3.87
PASST w/ soft align (T = 1)	18.89	0.22	2.97	0.37	3.82
PASST w/ soft align (T = 0.01)	18.76	0.23	2.99	0.37	3.89
PASST w/ word align	18.69	0.23	3.00	0.38	3.89

Table 2. S2ST results on mExpresso En-Es

Methods	ASR-BLEU ↑	S-Sim ↑	A.PCP ↑	NISQA ↑
Unit-to-Unit [5]	13.76	0.06	2.37	3.15
SeamlessM4T [22]	19.55	0.04	2.45	3.13
TransVIP [9]	23.37	0.19	2.68	3.23
PASST w/ hard align	25.50	0.24	2.92	3.70
PASST w/ soft align (T = 1)	25.72	0.23	2.87	3.72
PASST w/ soft align (T = 0.01)	25.70	0.23	2.89	3.75

During training, the speech semantic encoder remains frozen and only isochrony encoder, decoder and newly itialized token embeddings are updated. We train separate models for each translation direction (En-Fr and En-Es). Training is carried out on 8 NVIDIA A100 GPUs with a batch size per device of 8, for approximately one week per model. We utilize pre-trained FACodec model trained on LibriLight [25] dataset. During inference, beam-search decoding is applied with a beam size of 5. We have utilized the attention weights between final encoder outputs and 6th layer of the 12-layer transformer decoder with averaging over all attention-heads, which produced the best results.

4.3. Evaluation Metrics

ASR-BLEU: We evaluate semantic quality using the BLEU score between the ground-truth text and the ASR-transcribed output speech (via Whisper-large [19])).

Auto-PCP: We measure utterance-level prosody similarity between source and translated speech using the AutoPCP [23], an automatic similarity predictor trained on human-ratings for prosody similarity between utterances.

Aligned Prosody Similarity: We assess fine-grained prosody transfer by measuring the frame-level cosine similarity of features from a Masked Prosody Model (MPM) [26] between semantically aligned words. These word-level alignments are established using ASR timestamps and an MT aligner [20]. Prosody features of target word are interpolated to match source sequence length. This metric is not reported for En-Es because MT aligner does not support that direction.

Speaker Similarity: We measure speaker similarity between source and translated speech using speaker embeddings from a speaker recognition model [23].

Naturalness: We measure overall naturalness of translated speech using NISQA-TTS [27], an automatic naturalness predictor trained on human ratings.

5. RESULTS & DISCUSSION

As the baselines, we report the results for TransVIP [9], SeamlessM4T [22], and Unit-to-Unit [5] models. We use official releases for SeamlessM4T and Unit-to-Unit while we reproduced TransVIP using the the official implementation on the same training conditions with the proposed method.

Table 3. Effect of Acoustic Tokens

Methods	ASR-BLEU ↑	S-Sim ↑	A.PCP ↑	P-Sim ↑	NISQA ↑
PASST w/ aligned acoustic tokens	18.76	0.23	2.99	0.37	3.89
PASST w/ no acoustic tokens	18.60	0.23	2.97	0.37	3.87
PASST w/ predicted acoustic tokens	18.78	0.21	2.78	0.33	3.13

The results can be seen in Table 1 and Table 2. Firstly, the proposed method has the best translation performance in both cases, which can be attributed the superior disentanglement of content features potentially making semantic translation task easier. As expected, the proposed method PASST, outperforms all the baselines in prosody similarity metrics for both En-Fr and En-Es cases proving the effectiveness of direct prosody transfer approach. For speaker similarity, the SeamlessM4T and Unit-to-Unit baselines are not capable of preserving source speaker identity while proposed method outperforms similar TransVIP baseline, that can be attributed to the better preservation of prosody which is a part of speaker style. Finally, since modeling rich prosodic details is crucial to have natural-sounding speech, we believe better preservation of prosodic details reflects on naturalness as PASST has higher NISQA scores than almost all of the baselines. One-tailed paired t-tests validate statistical significance ($p < 0.05$) for best method vs second-best in every metric. We note that Unit-to-Unit can only model single-speaker which is an easier task and might led to slightly better naturalness in En-Fr.

Soft-alignment consistently yields slightly better performance, likely due to its smoothing effect, which makes it more robust to occasional attention noise. More importantly, our attention-based alignment performs on par with a strong text-based alignment approach, proving its robustness.

We investigated the necessity of acoustic token alignment, given their unclear role in speech reconstruction. Our experiments, in Table 3, where these tokens were omitted during synthesis had a negligible impact on translated speech. However, predicting these tokens caused a significant degradation in both prosody similarity and naturalness. This suggests that while original acoustic tokens from source may have limited effect, predicted acoustic tokens introduce errors that harms generation. This finding supports our paper's motivation: direct transfer of original source attributes might be a more robust and effective strategy than generating them from scratch.

6. CONCLUSION

We introduced a novel approach for preserving source prosody in speech-to-speech translation through direct prosody transfer with disentangled speech features. Unlike prior indirect methods, our approach explicitly aligns and transfers fine-grained prosodic details using attention maps from the content translator. Experiments show significant gains in prosody preservation, along with improvements in translation accuracy, speaker similarity, and naturalness. This higlights the importance of explicit prosody transfer in S2ST and opens directions for broader languages and transfer of emotions.

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